

Business Requirements Document - LogiMaxX

Business Requirement Document (BRD)

LogiMaxX (MVP1)

Client:

Maruti Suzuki India Limited (MSIL)

Department:

TBD

Date:

28-02-2024

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0.1

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Draft

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Change History

Version	Status	Date	Change	Chapter	Author
0.1	Draft	28-02-2024	Draft version	-	Maruti IoT Team

Document Approvals

Role	Name	Signature	Date

Project summary

Objectives

1. Trace the movement of the trucks within the plant locations.
2. Track the Entry time, Exit time and TTAT of trucks inside the plant location. Trucks can either do normal/single dock delivery or do a milk run/multi-dock delivery.
3. Track the time spent by trucks in various areas within the plant location.
4. Ability to view the bay availability status (Based on dock level Bay count for POC)
5. Notify guards of any alarms raised for a truck in various areas.
6. POC of the physical devices to determine if they can be deployed to other plant areas/locations.

Background

Maruti Suzuki India Limited (MSIL) is focused on excelling in the Sustainable Development Goals (SDGs) defined by United Nations. The Company strives to reduce the impact of its products and services on the environment through the sustainable use of natural resources and responsible waste, water, and emissions management practices. MSIL has increased the use of solar power and canal water for manufacturing with 62% of water requirement met through recycling. MSIL sent 86,000 MT of metallic scrap for recycling. It co-processed 11,569 MT of hazardous waste. The Company deployed CNG and smart hybrid technology in its products to minimize the carbon emissions. In 2017, the Company voluntarily started activities to ensure RRR (Reusability, Recyclability, Recoverability) compliance of all domestic models ahead of Indian regulations. In 11 products, the Company uses the IMDS (International Material Data System) tool to measure the 'RRR' performance. By 2022, the Company targets to implement IMDS tool in all its products (MSIL, 2021).

In MSIL's Manesar Powertrain Plant, spanning 1.8 square kilometers, close to 400 trucks transport consignments daily to support operations across various production sites, such as the K series engine plant, Transmission plant, and 2W engine plant. Truck movements are mainly synchronized with general shift hours to enable effective communication with suppliers. Also, truck movement is constrained between 14:30 and 15:30 to mitigate safety risks during shift changeovers. The material delivery takes place at specified unloading zones referred to locally as bays. Truck movements within the plant are categorized into procurement and dispatch activities. Procurement involves E-Nagare trucks, which adhere to scheduled delivery times, and DI and non-DI trucks, which operate based on different scheduling criteria. Dispatch activities entail the transportation of engines, transmissions, and waste materials to various destinations.

Despite ongoing operations, there are still critical challenges that remain unresolved. Foremost among them is the absence of monitoring for truck turn around time (TTAT), which poses safety and compliance risks. Furthermore, inefficient bay utilization and prolonged waiting times for trucks amplify delays in material deliveries, disproportionately affecting suppliers with limited business shares. Effectively addressing these challenges was needed to improve operational efficiency and fostering seamless truck movements within the plant premises.

The 'LogiMaxX' application solution will provide end users with the capability to monitor truck movements within the 'Manesar Powertrain Plant', particularly in the KB-KC plant area. Users will have access to real-time data concerning the total number of trucks present in the plant, their current locations, and additional relevant details. Additionally, users will view alarms in case a truck exceeds the target truck turn around time (TTAT), creates congestion, remains idle in plant areas, deviates from the ideal route, or experiences connectivity issues. Furthermore, users will have the ability to communicate with truck drivers using the contact information provided during the check-in process, and they will also be able to retrieve historical reports for analysis purposes.

Business Drivers

1. **Operational Efficiency:** Streamlining truck movements and logistics processes within the plant premises can significantly enhance operational efficiency. By ensuring synchronized truck movements with shift hours and efficient utilization of bays, the overall productivity of operations can be improved.
2. **Compliance and Safety:** Implementing monitoring systems for truck turn around time (TTAT) addresses safety and compliance concerns. This ensures that operations adhere to regulatory requirements and mitigate risks associated with prolonged truck wait times and congested areas.
3. **Supplier Relationship Management:** Improving truck movement efficiency reduces delays in material deliveries, particularly for suppliers with limited business shares. Enhancing these relationships through timely deliveries can positively impact supplier satisfaction and long-term partnerships.
4. **Real-time Visibility:** The 'LogiMaxX' application provides real-time visibility into truck movements, enabling better decision-making and resource allocation. Users can monitor truck locations, TTAT adherence, and potential issues such as congestion or deviations from optimal routes, allowing for proactive management.
5. **Data-Driven Analysis:** Access to historical reports through the 'LogiMaxX' application enables data-driven analysis of truck movements and logistics performance. This facilitates continuous improvement initiatives and strategic decision-making to optimize operations further.

Business Users

1. Plant Head
2. Division Head
3. Department Head
4. Department managers
5. Vigilance Department
6. Mx (Material Receiving Department)

Access Roles

The following access roles will be maintained at 'Location' level. For this MVP, the location we are taking is 'Manesar Powertrain Plant'.

- **Module access** – This role will provide the user access to view the information available in the dashboard and other screens of the solution. The user will not be able to carry out transactions such as master data uploading or downloading of csv(database) reports.
- **Master access** – This role will provide the user complete access to the solution. The user will be able to view the information, upload master data, download last uploaded master data, download csv(database) reports, and use any other added functionality.
- **Download access** – This role will provide the user access to view the information available in the solution as well as download csv reports wherever applicable.

Project Scope

In Scope

Please refer Functional Requirements section for software scope.

Hardware Scope:

S.No	Hardware Device	Quantity
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1	BLE Tags(15) + Magikk(10)	25
2	BLE LoRaWAN Nodes/ BLE Receivers	20
3	LoRaWAN Gateway	1

Out of Scope

- Contacting driver (move to bay, move outside, go to parking etc.) – To be manually done by security guard
- Informing driver of available bay – to be manually done by security guard.
- Integration with existing MSIL systems- To be considered in future MVPs.
- Device Monitoring.

System perspective

Assumptions

- Master file uploaded will contain the information in the correct template.
- Only KB-KC plant area at MSIL Powertrain Plant, Manesar is considered for current project scope.
- Only Dock level truck tracking is in scope.
- Bay level truck identification is out of scope.
- Only trucks delivering material to the KB-KC area will be given the BLE/GPS tags.
- The BLE/GPS tag mapped to a truck will be the only source of data to track the truck.
- Internet connection will be required to upload data from gateway to cloud and will be provided by MSIL.
- No integration is required with existing systems.
- In current project scope, LoRaWAN gateway will directly communicate with AWS cloud.

Dependencies

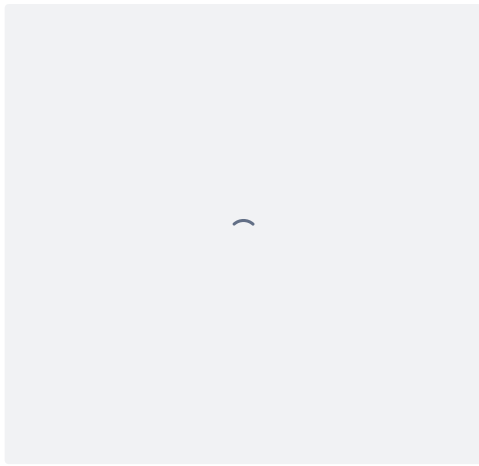
- MSIL will ensure that BLE/GPS tags are provided with readable QR codes.
- MSIL will ensure Wi-Fi connectivity to upload data from gateway to cloud and will be provided by MSIL.
- MSIL will ensure battery charging of the BLE/GPS tags.

Risk

- Security guard not willing to check-in/out the trucks as per process.
- Business users not using the software application.
- Any hardware related issues encountered.

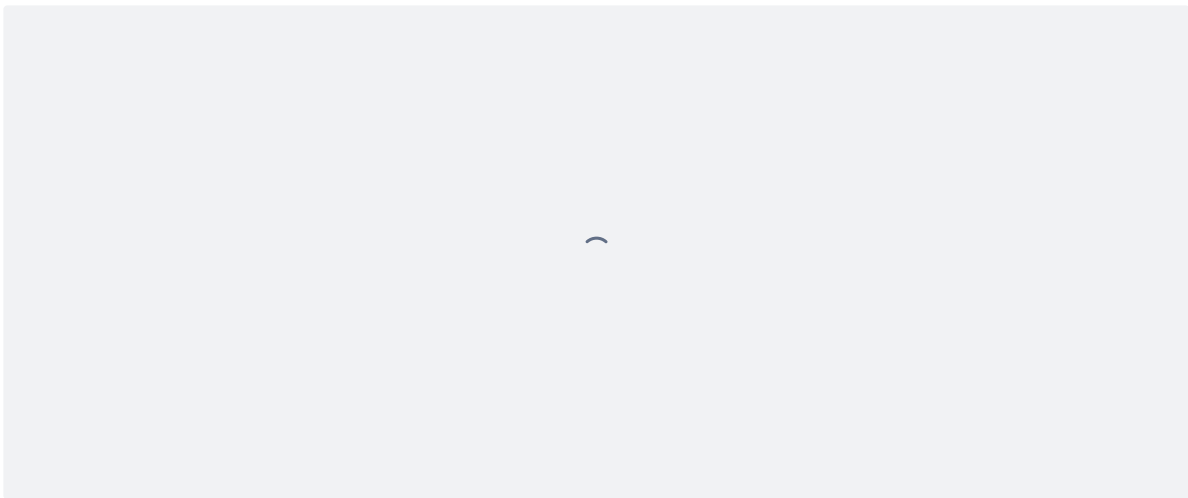
Business Process Overview

The Manesar Powertrain Plant area layout is as given below. Only KB-KC plant area is considered in-scope for current project implementation.



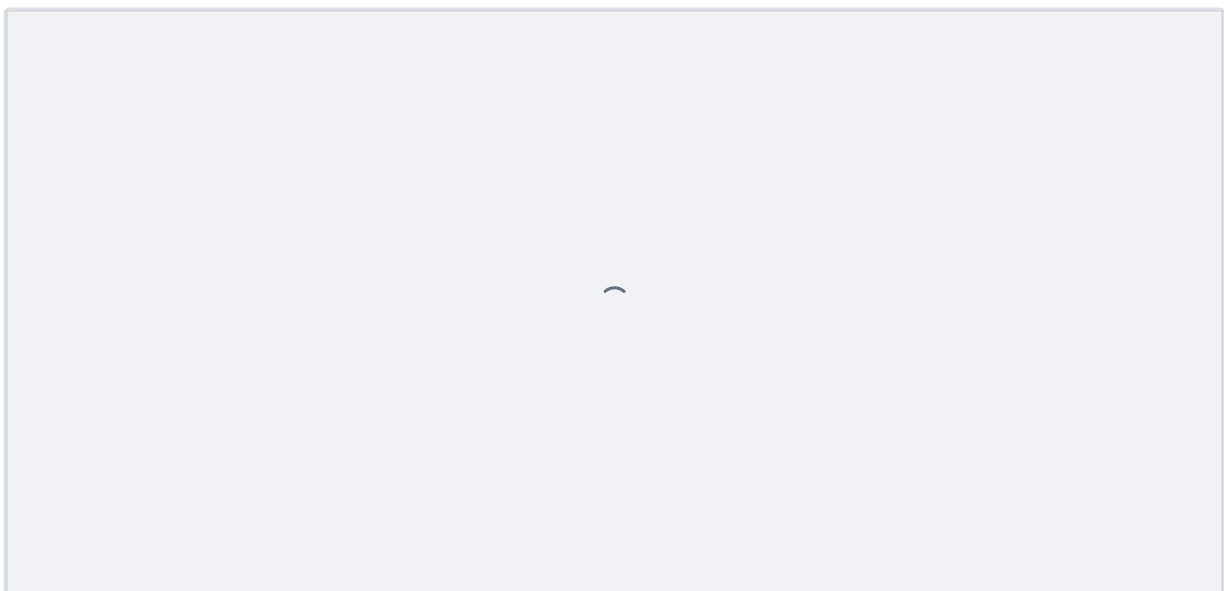
CURRENT PROCESS

The existing process of truck entry and material delivery is as given below:



CURRENT PROCESS

FUTURE PROCESS



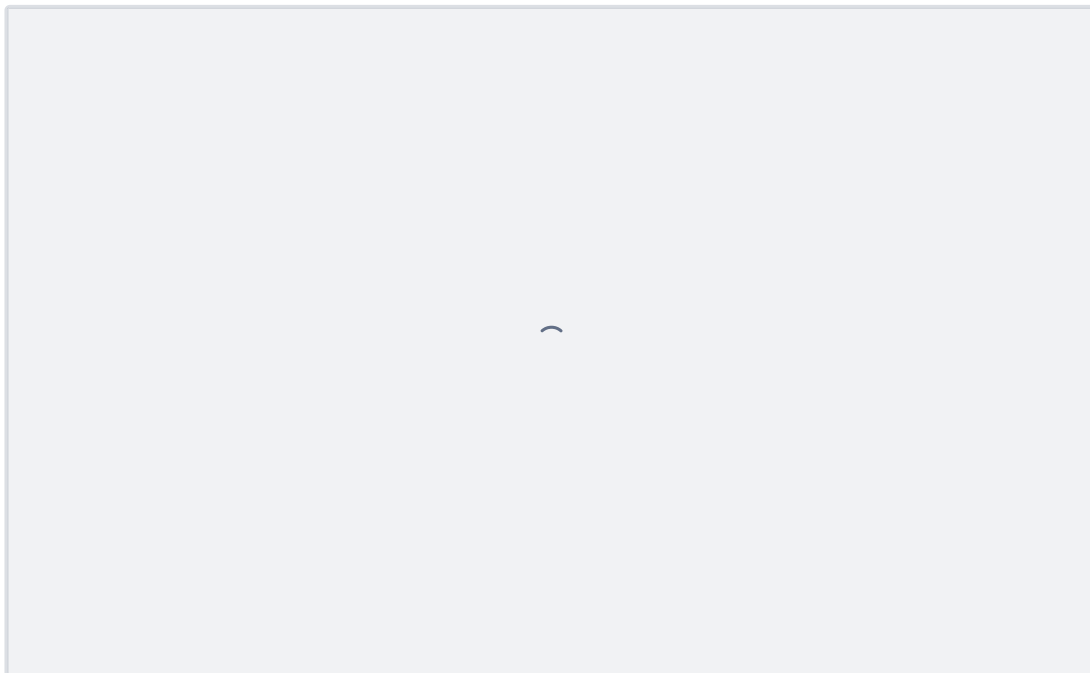
FUTURE PROCESS

Mockups

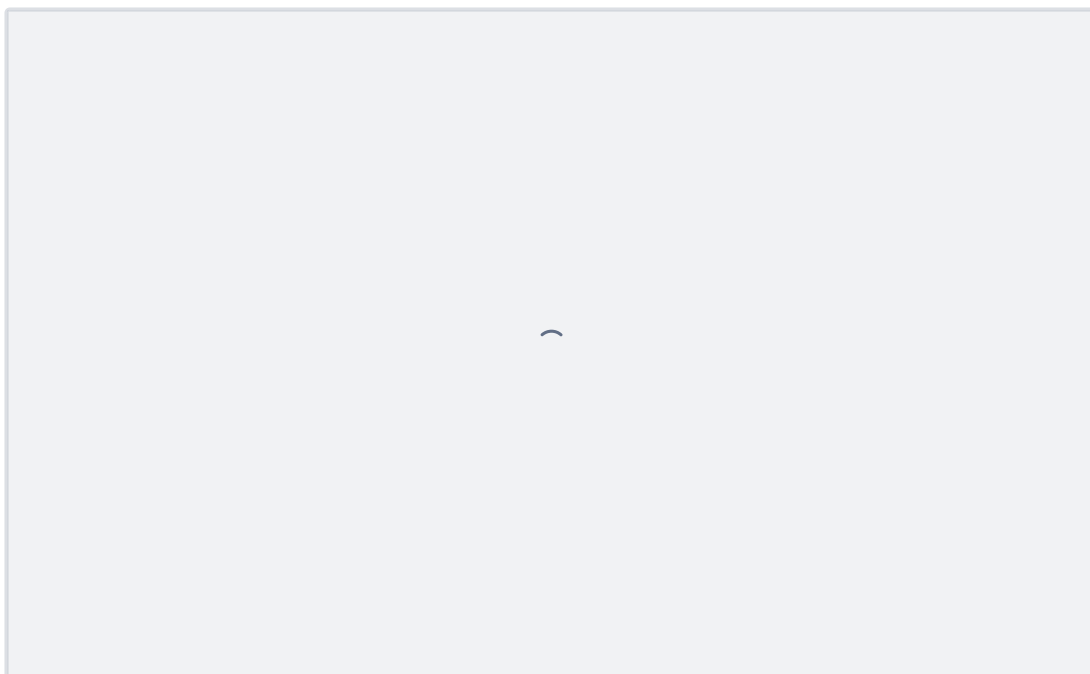
The following link can be used for viewing the latest system mockups:

✓ [TiTo - MSIL IOT - TiTo \(figma.com\)](#)

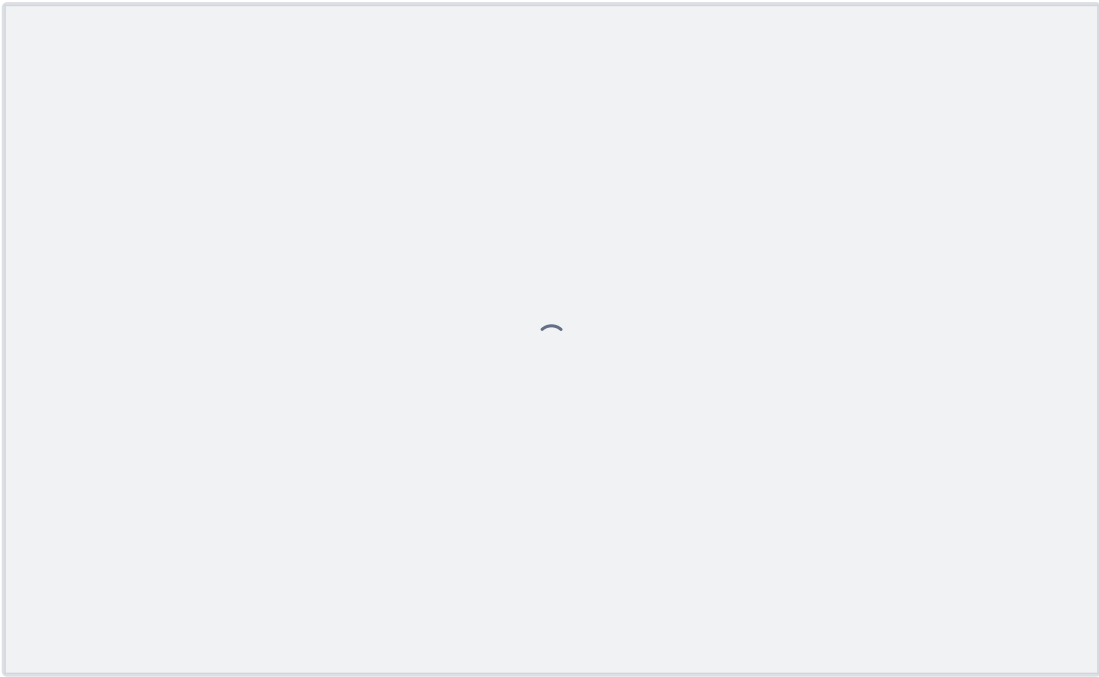
LogiMaxX Web Application



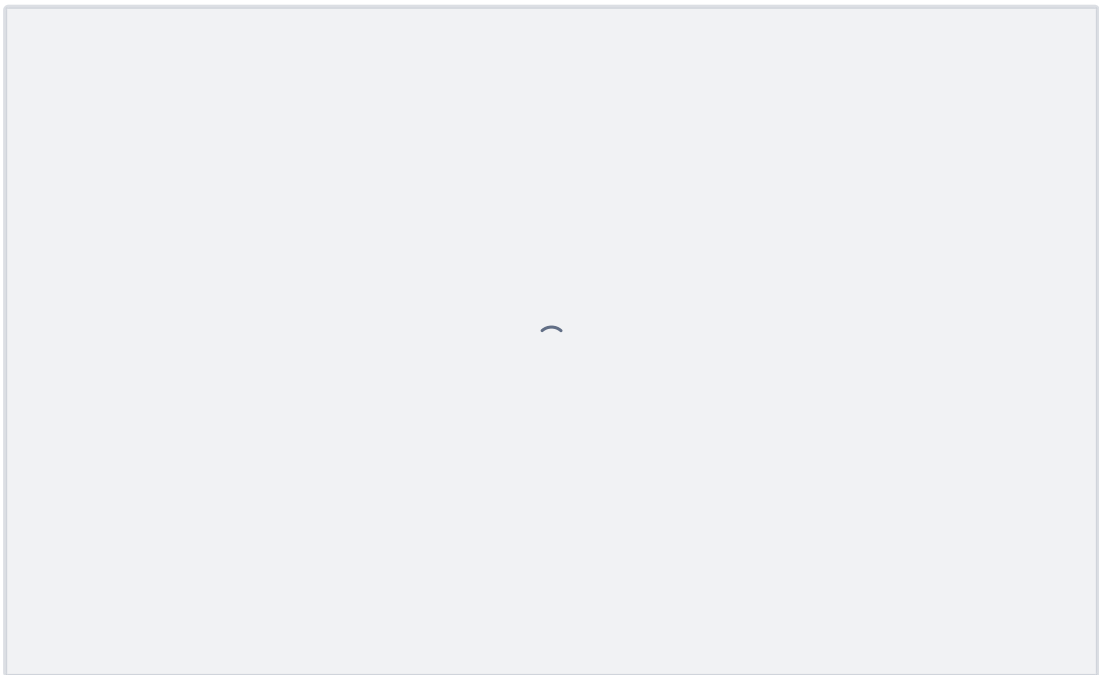
Dashboard - All Map View



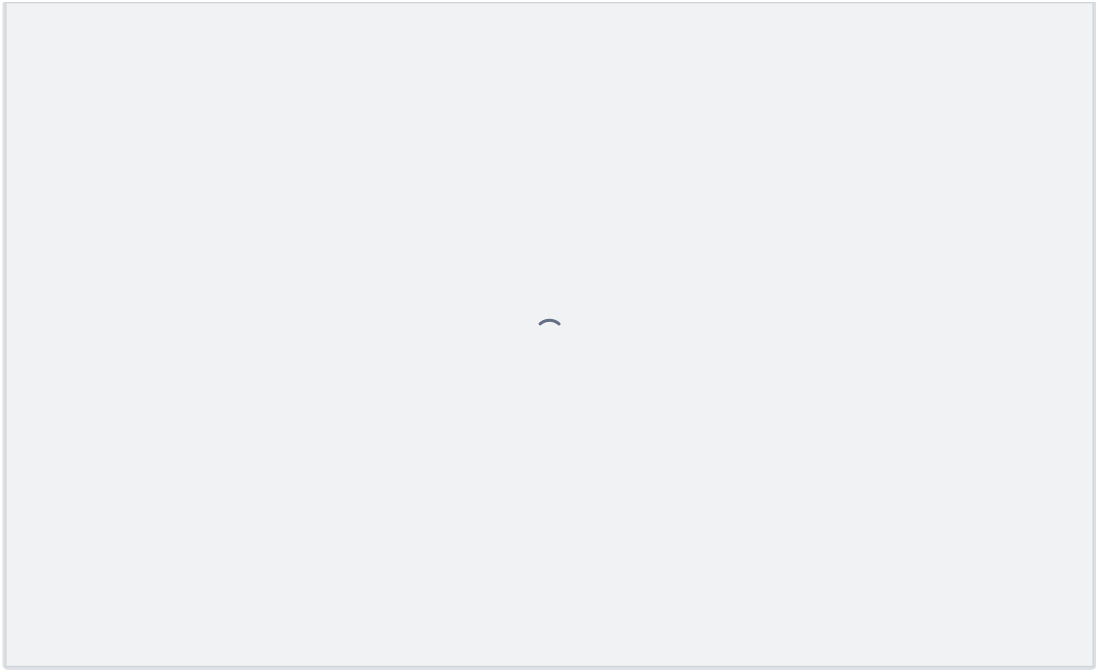
Dashboard - KB-KC Plant Map View



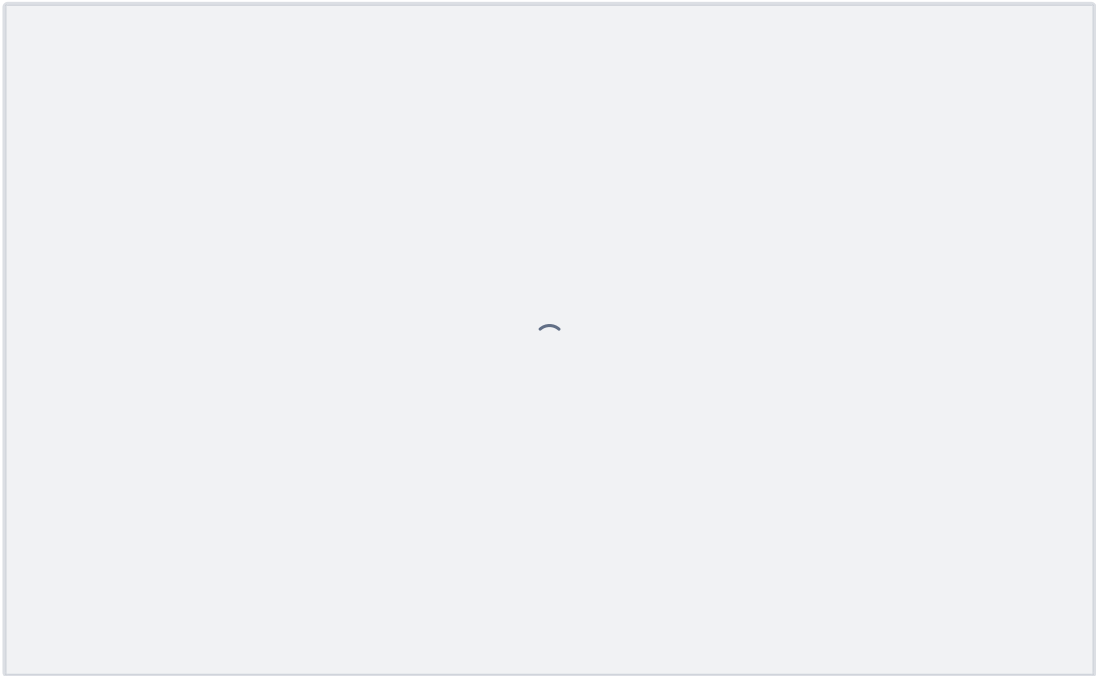
Dashboard - Truck Card Details



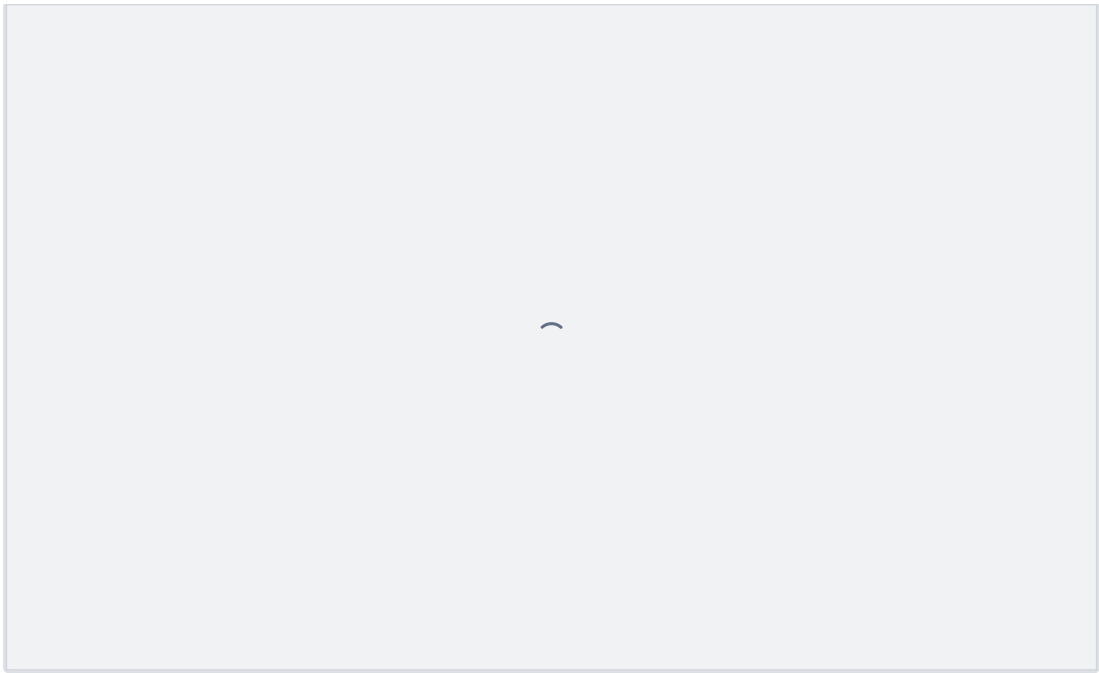
Dashboard - Search result view



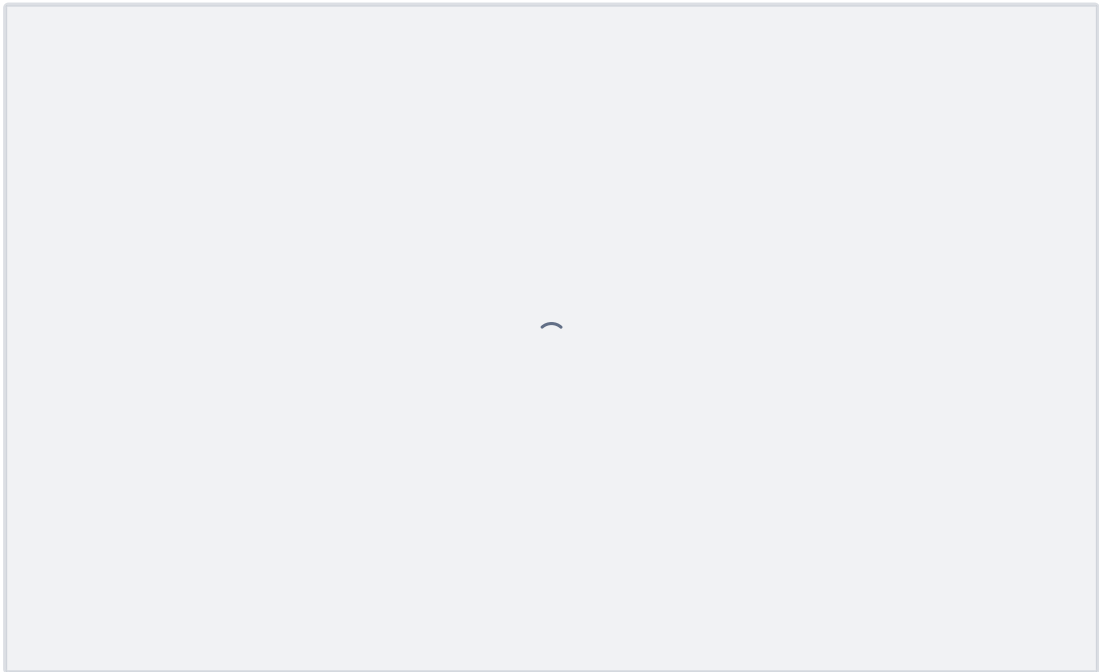
Dashboard - Alarm tab View



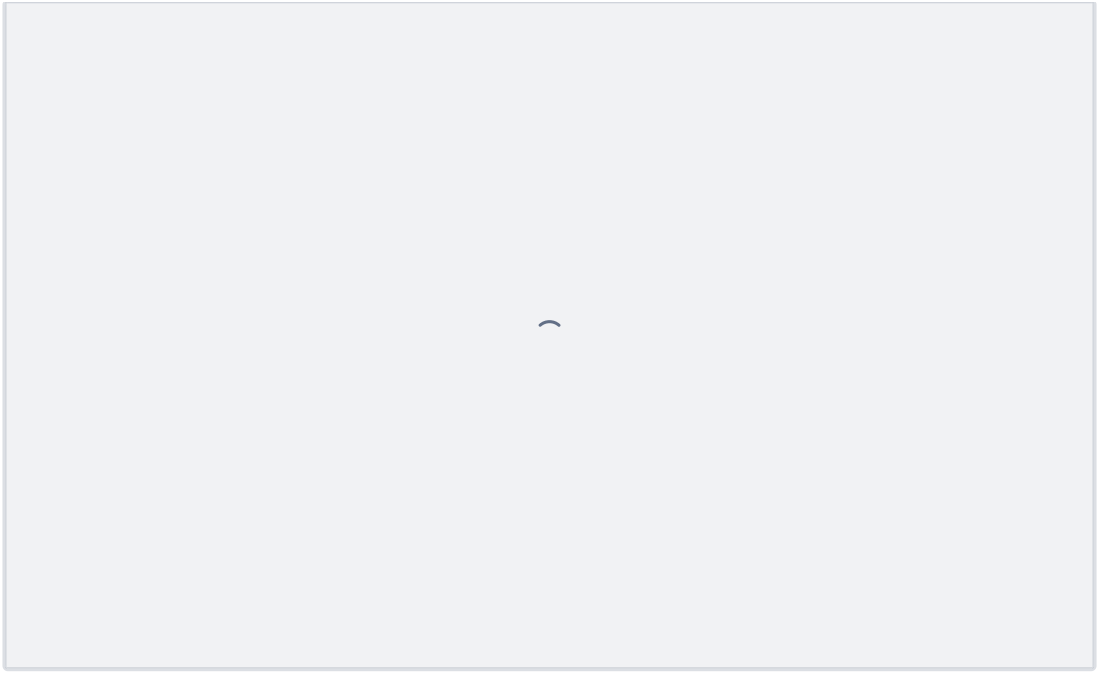
Dock Analysis - Live View



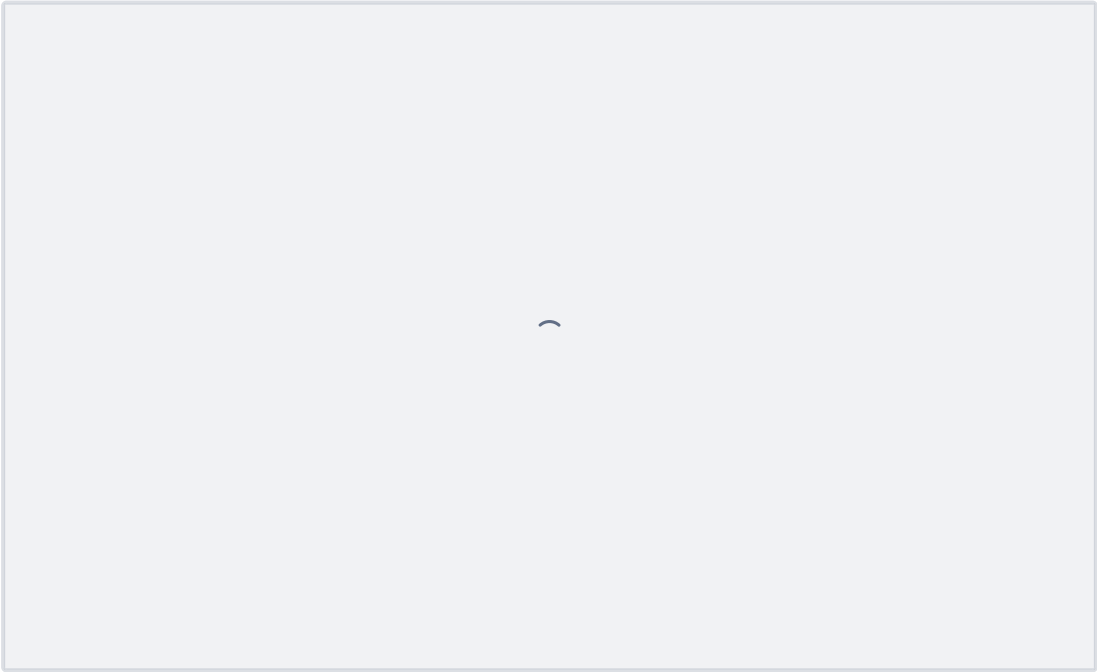
Dock Analysis - Historic View



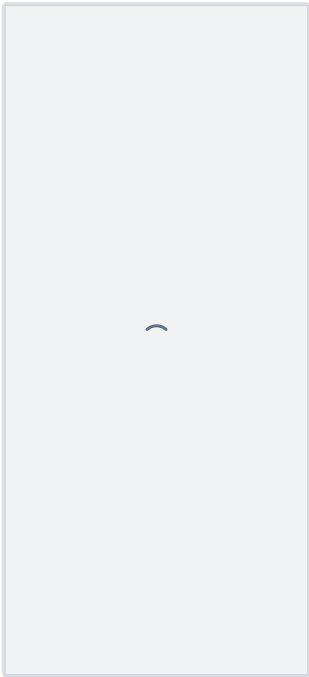
TTAT Analysis - Live View



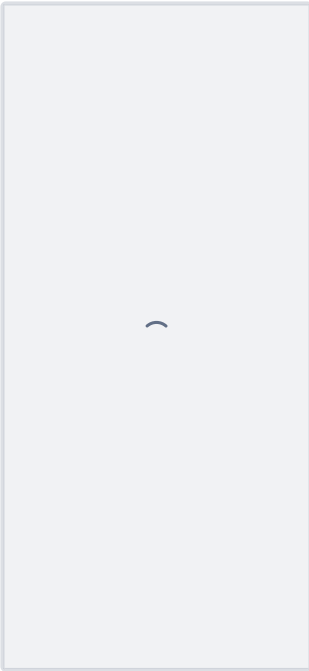
TTAT Analysis - Historic View



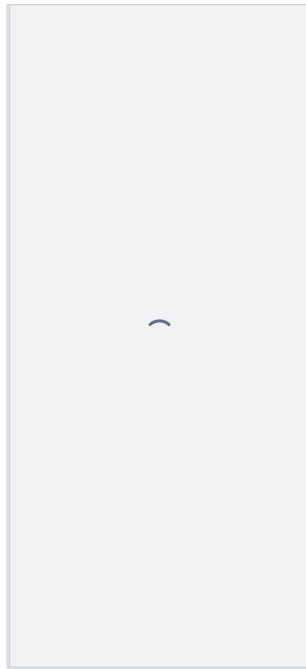
Alarm Report - Live & Historic View



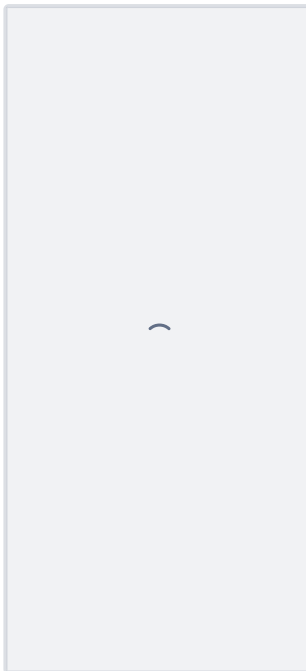
Landing Page



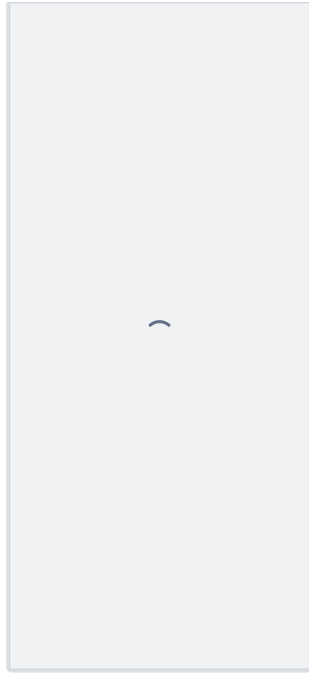
Tag Check In - QR Code scan



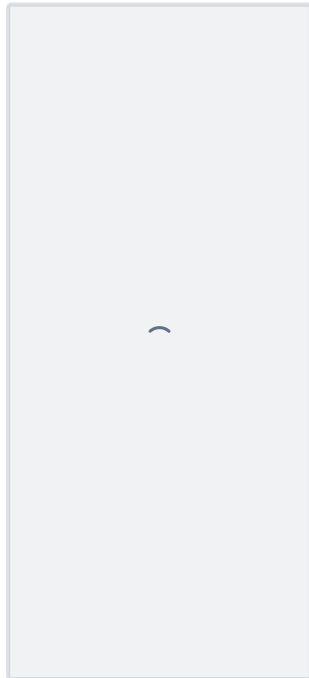
Scan Consignment No. - Bar
Code scan



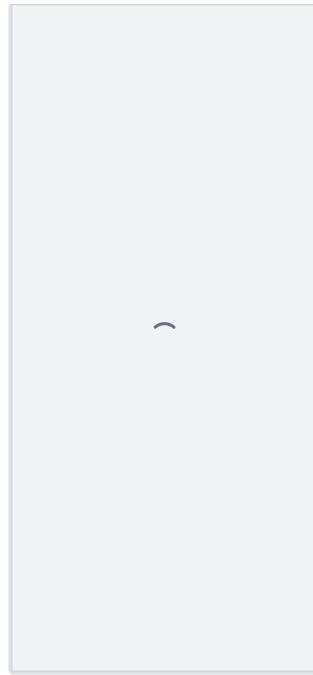
Tag Check In Details



Check In Confirmed



Tag Check Out - QR Code
scan



Check Out Confirmation

Use Cases

<TBD>

Master Data

The master data need to be maintained for the following:

- Target TAT for Plant Location
- Plant Areas within a Plant Location
- Docks within each Plant Area and their relevant details
- Zones defined within a Plant Location and their relevant details
- Ideal Route for trucks entering a Plant Location
- Mobile number associated with GPS/GPS+BLE tags.

Business Requirements

The requirements in this document are scoped as follows:

Value	Description
MVP1.0	This requirement is going to be released in MVP1.0

MVP2.0	This requirement is going to be released in MVP2.0
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Functional Requirements

Req #	Scope	Description	Rationale	Screen
General/ Base Functionality				
FR-G-01	MVP1.0	Dashboard tab as the default landing page/Home page for users	Live view	LogiMaxX App-Dashboard
FR-G-02	MVP1.0	Display the current production day truck movement as default view.	Live view	LogiMaxX App-Dashboard
FR-G-03	MVP1.0	Display the total count of trucks with BLE/GPS tags that are inside the Plant.	Live view	LogiMaxX App-Dashboard
FR-G-04	MVP1.0	In Map view, display the count of trucks within each dock/zones in Map view.	Live view	LogiMaxX App-Dashboard
FR-G-05	MVP1.0	In Map view, display the count of alarms associated with trucks within each dock/zones.	Truck Movement Monitoring	LogiMaxX App-Dashboard
FR-G-06	MVP1.0	In Map view, display the total bays occupied/available for each dock.	Dock/Bay Availability Monitoring	LogiMaxX App-Dashboard
FR-G-07	MVP1.0	Provide a filter 'Plant Area' to view the map area at plant or plant area level.	Specific View	LogiMaxX App-Dashboard
FR-G-08	MVP1.0	Display Map view at Plant level and Plant Area levels.	Specific View	LogiMaxX App-Dashboard
FR-G-09	MVP1.0	In Truck TAB view, display list of Trucks with BLE/GPS tags and their details.	Truck Monitoring	LogiMaxX App-Dashboard
FR-G-10	MVP1.0	For each truck, display : - Truck Registration Number - Supplier Name - Contact Numbers - Material Gate Entry Time - Elapsed Time (Time spent till now inside the plant) - Driver Name - BLE/GPS Tag ID - Current Location - Consignment No.	Truck Movement Monitoring	LogiMaxX App-Dashboard

		- Route Details (Entry-Exit time zone)- show journey of truck as the truck moves from one zone to another - Alarms		
FR-G-11	MVP1.0	Show the active alarms for a truck. Alarms can be of below five types: 1. Delay 2. Idle 3. Congestion 4. No Zone 5. Wrong Route	Show alarms	LogiMaxX App-Dashboard
FR-G-12	MVP1.0	In Alarm TAB view, display list of Trucks having only alarms.	Truck Movement Monitoring	LogiMaxX App-Dashboard
FR-G-13	MVP1.0	On both TABS, provide a filter 'Occupancy Area' to filter the list of trucks by zone/dock.	Zone/Dock Specific View	LogiMaxX App-Dashboard
FR-G-14	MVP1.0	Provide an auto-suggest Search Box to filter the list of trucks on both tabs by the following attributes: - Driver Name - Registration Number - Phone Number - Consignment Number - Vendor Name	Search	LogiMaxX App-Dashboard
FR-G-15	MVP1.0	Provide an option to filter the list of trucks on TAB view by clicking on zone cards/truck counts/alarm counts on Map View.	Zone/Dock Specific View	LogiMaxX App-Dashboard
FR-G-16	MVP1.0	Master screen with upload master file functionality and table to view file upload, approval/rejection details (reviewer, review date & time, submission date & time)	Upload Master file	LogiMaxX App-Master
FR-G-17	MVP1.0	On Master screen, provide error details on info 'i' icon under status to view error details when file upload is failed.	View file upload error details	LogiMaxX App-Master
FR-G-18	MVP1.0	On Master screen, provide functionality to download uploaded master files.	Download Master file	LogiMaxX App-Master
FR-G-19	MVP1.0	Provide auto-refresh after every 30 seconds interval on Dashbaord screen.	Auto Refresh	LogiMaxX App-Dashboard
FR-G-20	MVP1.0	Provide a refresh button on the Report and Master screens for users to view the latest data.	Manual refresh	LogiMaxX App-Master

FR-G-21	MVP2.0	'REPLAY' tab to view historical snapshot of truck movement inside the plant during a specific time period.	Time specific Analysis	LogiMaxX App-Replay
FR-G-22	MVP2.0	View real-time truck movement of a truck with GPS tag from Point A (Outside Plant) to Point B (Inside Plant) on Map.	GPS tracking	LogiMaxX App-Dashboard
FR-G-23	MVP2.0	Extension of the functionality from one Plant Area (KB-KC) to remaining plant areas within the 'Manesar Powertrain' plant or to other plant locations.	Future Implementation	LogiMaxX App-All
FR-G-24	MVP2.0	Integration with E-Nagare and other MSIL systems.	Integration	LogiMaxX App-All
FR-G-25	MVP1.0	A landing/home page to link/unlink a BLE/GPS tag to a truck/driver at the Material entry gate. Two options on the page: • Tag Check In • Tag Check Out	Interface to link/unlink BLE/GPS tag to a truck/driver	LogimaxXInOut-Home
FR-G-26	MVP1.0	Provide Tag Check In functionality : 1. Scan QR Code on BLE/GPS tag 2. Scan Consignment No. on consignment slip of the incoming truck/driver 3. Provide Driver Name and Contact 4. Check In the Truck	Check In Truck	LogimaxXInOut-Check In
FR-G-27	MVP1.0	Provide Tag Out functionality: 1. Scan QR Code on BLE/GPS tag 2. View Associated Details 3. Check Out the truck	Check Out Truck	LogimaxXInOut-Check Out
FR-G-28	MVP2.0	Provide option to manually enter the BLE/GPS tag number.	Alternate option if Tag QR code is not scanning due to some reason.	LogimaxXInOut-Check In
FR-G-29	MVP2.0	Provide option to manually enter the Consignment number.	Alternate option if Consignment Bar code is not scanning due to some reason.	LogimaxXInOut-Check In
Reporting Requirements				
FR-R-01	MVP 1.0	Three reports to be provided: 1. Dock Analysis 2. TTAT Analysis 3. Alaram Report	Reporting requirement	Reports

FR-R-02	MVP 1.0	Provide a 'Dock Analysis' report to view the 'No. of Trucks serviced by Docks' based on live as well as historical data.	Analyze Dock Performance	Reports->Dock Analysis
FR-R-03	MVP 1.0	Filters: Plant Area, Docks, Live/Historic, From and To date filter for Historic	Plant Area, Dock, Date specific analysis	Reports->Dock Analysis
FR-R-04	MVP 1.0	Default (Live): Top 10 docks for current production day.	Analyze Dock Performance	Reports->Dock Analysis
FR-R-05	MVP 1.0	Historic View: Top 5 docks for last five days.	Analyze Dock Performance	Reports->Dock Analysis
FR-R-06	MVP 1.0	Historic View: Highlight same Dock on different days when user hovers over a dock bar on the charts.	Analyze Dock Performance	Reports->Dock Analysis
FR-R-07	MVP 1.0	Provide a 'TTAT Analysis' report to view the 'Truck Turn Around Time (TTAT)' by Trucks based on live as well as historical data.	Analyze TTAT for trucks	Reports->TTAT Analysis
FR-R-08	MVP 1.0	Filters: Occupancy Area, Trucks, Live/Historic, From and To date filter for Historic.	Plant Area, Dock, Date specific analysis	Reports->TTAT Analysis
FR-R-09	MVP 1.0	Default (Live): Top 10 trucks by TTAT for current production day.	TTAT specific analysis	Reports->TTAT Analysis
FR-R-10	MVP 1.0	Historic View: Top 5 trucks by TTAT for last five days.	TTAT specific analysis	Reports->TTAT Analysis
FR-R-11	MVP 1.0	Show truck details for a truck when user selects on the charts.	View truck details	Reports->TTAT Analysis
FR-R-12	MVP 1.0	Highlight same truck nos. when user hovers over a truck bar on the charts.	View truck details	Reports->TTAT Analysis
FR-R-13	MVP 1.0	Provide an 'Alarm Report' report to view the 'No. of alarms raised by Occupancy Area/Supplier' based on live as well as historical data.	Analyze Alarms generated	Reports->Alarm Report
FR-R-14	MVP 1.0	Filters : View by (Occupancy Area/Supplier), Occupancy Area/Supplier, Live/Historic, From and To date filter for Historic	Plant Area, Dock, Date specific analysis	Reports->Alarm Report
FR-R-15	MVP 1.0	Default (Live): Top 10 occupancy areas with highest no. of alarms raised for current production day	Analyze Dock Performance	Reports->Alarm Report
FR-R-16	MVP 1.0	Historic view : Top 5 occupancy areas with highest no. of alarms raised for last 5 days.	Analyze Dock Performance	Reports->Alarm Report
FR-R-17	MVP 1.0	Provide a refresh button on the screen for users to view the latest data.	Manual refresh	Reports->All
FR-R-18	MVP 1.0	Provide a download button on the screen for users to get the latest report download	Report Download	Reports->All

		on their local machine.		
Data Requirements				
FR-D-001	MVP1.0	Maintain master data for location and its Target TAT	Data Input	Master
FR-D-002	MVP1.0	Maintain master data for 'Plant Areas' within a location	Data Input	Master
FR-D-003	MVP1.0	Maintain master data for docks and bays within Plant Areas	Data Input	Master
FR-D-004	MVP1.0	Maintain master data for defined zones within a plant area	Data Input	Master
FR-D-005	MVP1.0	Maintain master data for Ideal Route	Data Input	Master
FR-D-006	MVP1.0	Maintain master data for Contact Numbers of GPS/GPS+BLE Tags	Data Input	Master
FR-D-007	MVP1.0	Parsing and validations related to master file	Data Validation	Master
Security Requirements				
FR-S-01	MVP1.0	Create 'View' access role	System Security	Master
FR-S-02	MVP1.0	Create 'Master' access role	System Security	Master
FR-S-03	MVP1.0	Create 'Download' access role	System Security	Master
FR-S-04	MVP1.0	Configure workflow for approval of master files	Ensure Data Correctness	Master

Non-Functional Requirements

ID	Requirement
NFR-001	Availability on webapp
NFR-002	Performance
NFR-003	Scalability
NFR-004	Data quality
NFR-005	Less clicks with more information

Appendix

Acronyms

Term	Definition
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BRD	Business Requirement Document
TTAT	Truck Turn Around Time
POC	Proof Of Concept
BLE	Bluetooth Low Energy
GPS	Global Positioning System
LoRaWAN	Long Range Wide Area Network

Glossary

Term	Definition